Aditya Bhadauria

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Education

Graphic Era University Aug 2019 –May 2023
B.E. in Computer Science | CGPA: 8.0/10 Dehradun, India

Work Experience

Backgammon Galaxy - Tech(FullStack) Sept 2023 – Present
Senior Software Engineer Remote(Spain,Europe)

- **Built Computer Vision Match Transcription**
  - Built an end-to-end **video-to-transcript pipeline** turning match recordings into analysis-ready games.
  - Fine-tuned **RF-DETR** and YOLO models to detect the board, checkers and read dice values in the game.
  - Swapped per-frame dice inference for a **temporal transformer** over frame windows, handling blur and occlusion.
  - Architected a hybrid **GPU/CPU** pipeline processing **1,000+ matches daily** at **65%** lower GPU cost.
  - Added cross-frame **trajectory tracking** and automated move validation for consistent transcripts .
- **Developed Backgammon's Backend Architecture:**
  - Deployed and optimized the platform using **Kubernetes** and **Docker**, ensuring smooth performance and **99.9% uptime**.
  - Revamped **back-end infrastructure** with **asynchronous calls** and optimized system architecture, reducing API response time from over **1 minute to under 15 seconds**.
  - Configured **Kubernetes Horizontal Pod Autoscaler (HPA)** to handle sudden traffic spikes during peak tournaments.
  - Set up **Prometheus** and **Grafana** dashboards to track the health of the application and load of the server.
  - Built a Game Analysis Engine with **gRPC and RabbitMQ** to detect mistakes in real time for improving gameplay .
- **Integrated AI-Powered Game Features**
  - Integrated the **XG Gammon engine** to provide an AI opponent, allowing players to practice instantly.
  - Developed **AI-powered move suggestions** using **gnubg**, helping players analyze their matches and improve strategy.
  - Designed an **AI coaching system** that explains why players make mistakes using game rules and strategy knowledge.

Amplytic(Antler,2025) - Tech(Backend) Nov 2025 – Feb 2026
Software consultant Remote(London,United Kingdom)

- Built an **AI agent for asset managers** to automate workflows and improve investment returns.
- Architected and deployed multi-country **ETL Data pipelines** on GCP Cloud Run, processing **500K+** daily data points.
- Developed an **agentic workflow** that transforms raw reports into verified searchable knowledge using multi-step reasoning.
- Built a **hierarchical evaluation framework (NDCG)** to benchmark RAG performance and **improve search accuracy.**
- Implemented a **high-concurrency FastAPI backend** with **sub-100ms P99 latency**.
- Built a **Neo4j knowledge graph** to represent complex energy regulations, defining clear relationship types to accurately connect rules, entities, and compliance requirements, enabling efficient regulatory queries and analysis.

Nineleaps - Tech(FullStack) Dec 2022 – Sept 2023
Software Developer,Full time Bangalore, India

- Built a **smart water management dashboard** using **React, Node.js, PostgreSQL**, allowing city operators to track live water usage, detect leaks, and generate automated reports; reduced issue detection time by **20%**.
- Designed and implemented custom **D3.js charts** to visualize real-time consumption patterns with interactive filtering.
- Developed automated **alert and notification system** that informed operators within minutes of abnormal usage patterns.
- Optimized **CI/CD pipelines** using GitHub Actions and Jenkins for seamless build, test, and deployment workflows.
- Developed and maintained backend APIs and frontend components to support real-time data visualization and reporting for smart water management systems.

Projects and open source contributions

Nodeflow (GitHub)
- Built a **graph-powered code intelligence engine** that indexes any codebase into a knowledge graph.

AI Fashion Platform
- Built an **AI-powered fashion platform** that curates personalized outfits using recommendation systems (link).

Open Source Contributions
- Contributed to **Onlook (YC S25)** and built **evaluation benchmarks for AI agents** with **Datacurve.ai**.

Technical Skills

Languages:	Python, TypeScript, JavaScript, C/C++, Java, Rust, SQL
Frameworks:	React.js, Next.js, Node.js, FastAPI, Material UI
ML & Vision:	PyTorch, RF-DETR, YOLO, Transformers, OpenCV, RAG
Databases:	PostgreSQL, MySQL, MongoDB, Neo4j, Prisma
DevOps:	Kubernetes, Docker, Helm, AWS, GCP, CI/CD, Prometheus, Grafana
Tools:	Git, RabbitMQ, gRPC, Linux, Sentry, pytest

Interests

Led school football squad to compete at state level, enjoy board games and esports recreationally.